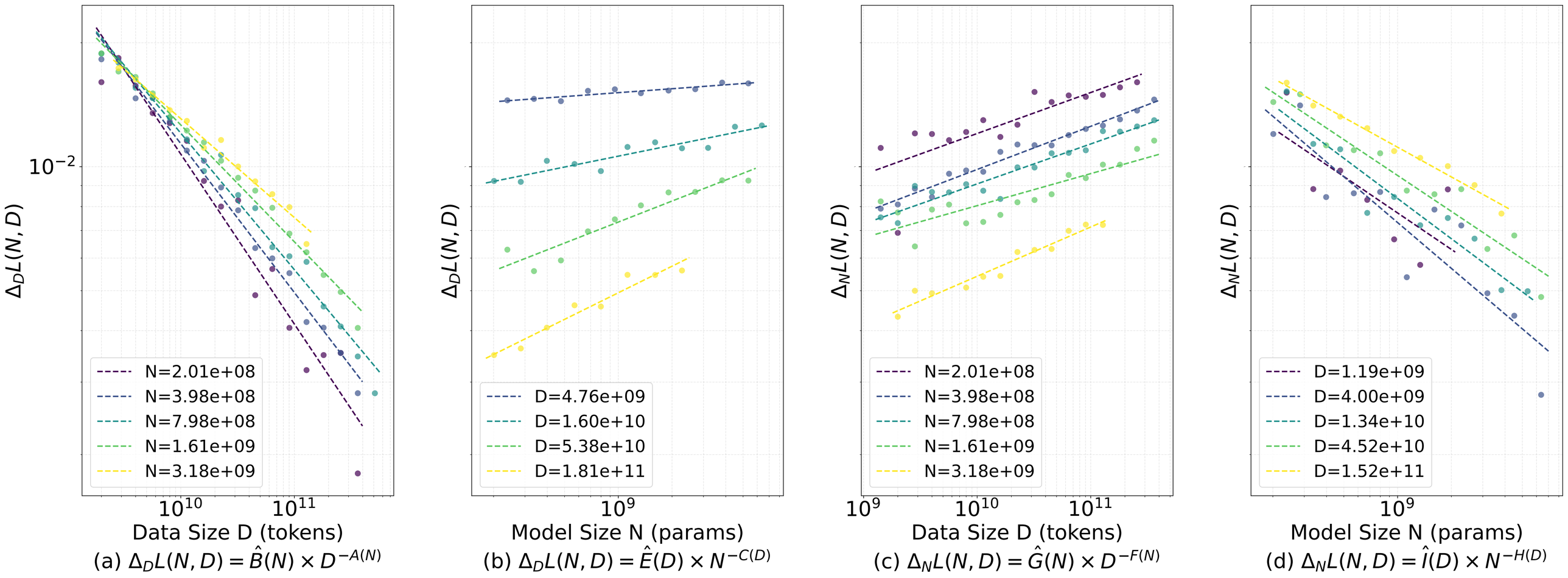
10^{-2}
$\Delta_D L(N, D)$
N=2.01e+08
N=3.98e+08
N=7.98e+08
N=1.61e+09
N=3.18e+09
10^{10}
10^{11}
Data Size D (tokens)
(a) $\Delta_D L(N, D) = \hat{B}(N) \times D^{-A(N)}$
$\Delta_D L(N, D)$
D=4.76e+09
D=1.60e+10
D=5.38e+10
D=1.81e+11
10^{9}
Model Size N (params)
(b) $\Delta_D L(N, D) = \hat{E}(D) \times N^{-C(D)}$
$\Delta_N L(N, D)$
N=2.01e+08
N=3.98e+08
N=7.98e+08
N=1.61e+09
N=3.18e+09
10^{9}
10^{10}
10^{11}
Data Size D (tokens)
(c) $\Delta_N L(N, D) = \hat{G}(N) \times D^{-F(N)}$
$\Delta_N L(N, D)$
D=1.19e+09
D=4.00e+09
D=1.34e+10
D=4.52e+10
D=1.52e+11
10^{9}
Model Size N (params)
(d) $\Delta_N L(N, D) = \hat{I}(D) \times N^{-H(D)}$